7 Let $\{f_n\}$ be a sequence in $L^1(X,\mathcal{M},\mu)$ such that $f_n \to f$ uniformly, $\mu(X) < \infty$. We want to show that $f \in L^1$ and $\lim \|f - f_n\|_1 = 0$. Note that for some $\epsilon > 0$ there exists N_ϵ such that for all $n \geq N_\epsilon$ we have that $|f - f_n| < \frac{\epsilon}{\mu(X)}$. Therefore $\int_X |f - f_n| d\mu < \frac{\epsilon}{\mu(X)} \int_x d\mu = \epsilon$. Therefore in the limit it goes to zero. Additionally $\int_X |f| d\mu \leq \int_X |f - f_n| d\mu + \int_X |f_n| d\mu < \infty + \epsilon = \infty$, giving us that $f \in L^1$. Note that this fails to hold if $\mu(X) = \infty$. In particular if one takes the functions $f_n(x) = \sum_{k=1}^\infty \frac{1}{k^{1+\frac{1}{n}}} 1_{(k-1,k)}(x)$, each $\int_{\mathbb{R}} |f_n| d\mu$ is guarenteed to converge by the p-series tests. However if one takes n to infinity you get the harmonic series, which diverges.

8 Let $f(x) = x^{-1/2} 1_{(0,1)}$, $\{q_n\}_{n\in\mathbb{N}}$ be an enumeration of the rationals, and $g(x) = \sum_{n=1}^\infty f(x - q_n)$. We claim that $g \in L^1$. Note that first each $2^{-n} f(x - q_n)$ is an integrable function, therefore we can interchange the sum and the integral: $\int_{\mathbb{R}} |g| d\mu = \int_{\mathbb{R}} \sum_{n=1}^\infty 2^{-n} f(x - q_n) d\mu = \sum_{n=1}^\infty \int_{\mathbb{R}} 2^{-n} f(x - q_n) d\mu$. Furthermore, by the construction of f, $\int_{\mathbb{R}} f(x - q_n) d\mu = \int_{\mathbb{R}} f d\mu$, since the shifting of both the function and the indidcator exactly cancel out. Therefore $\int_{\mathbb{R}} |g| = \sum_{n=1}^\infty 2^{-n} \int_{\mathbb{R}} f d\mu = \sum_{n=1}^\infty 2^{1-n} = 1$. We additionally claim for all intervals $I \subset \mathbb{R}, \lambda \geq 1$ that $\mu(I \cap \{x : g(x) > \lambda\}) > 0$. Let $(a,b) = I, \lambda \geq 1$, and let q_a, q_b be rational numbers such that $a < q_a < q_b < b$ and $q_b - q_a < 1$. If we then consider $2^{-\alpha} f(x - \frac{q_b + q_a}{2})$ where α is the index of $\frac{q_b+q_a}{2}$ we are guaranteed that g has a singularity contained within said interval. Furthermore, the interval $(\frac{q_b+q_a}{2}, \min(\frac{1}{\frac{q_b+q_a}{2} + \lambda^2}, q_b)$ is guaranteed to be of non-zero measure. Thus $\mu(I \cap \{x : g(x) > \lambda\}) > 0$.

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13 Let $(X, \mathcal{M}, \mu)$ be a measure space, μ is a semifinite measure, f is a non-negative function measurable function such that for all $g \in L^+(X, \mathcal{M}, \mu)$,

$$\int_X fg d\mu < \infty.$$

We claim that f is uniformly bounded almost everywhere. Suppose not. Then for all $M > 0$ there exists a set E_M such that for all $x \in E_M, f(x) > M$